

Set's in Calculus

Kovas McCann

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1 Introduction

To “compile” this information I am using the book Calculus by David Patrick it can be found here: <https://artofproblemsolving.com/store/book/calculus>.

This Document is designed to review the Set Theory principals covered in the aforementioned book. I will create other “Guides” for other topics covered. They can be found here: <https://github.com/KovasMcCann/Writings>.

2 Definition (What is a Set)

A Set is a collection of objects

- Objects can only be in the set once
- Objects can be anything “quantifiable”
- Order in a Set doesn't matter

3 Types of Sets

1. Finite Set

- A set with a finite number of elements.
- Example: $S = \{1, 2, 3\}$

2. Infinite Set

- A set with an infinite number of elements.
- Example: $S = \{1, 2, 3, \dots\}$

3. Subset

- A set A is a subset of set B if every element of A is also an element of B .

- Example: If $B = \{1, 2, 3\}$, then $A = \{1, 2\}$ is a subset of B ($A \subseteq B$).

4. Superset

- A set B is a superset of set A if every element of A is also an element of B .
- Example: If $A = \{1\}$ and $B = \{1, 2, 3\}$, then B is a superset of A ($B \supseteq A$).

5. Proper vs. Improper Subsets and Supersets

- A set A is a **subset** of set B if every element of A is also an element of B . This is written as $A \subseteq B$.
- A set A is a **proper subset** of set B if $A \subseteq B$ and $A \neq B$. This is written as $A \subset B$.
- **Example:** If $A = \{1\}$ and $B = \{1, 2, 3\}$, then A is a proper subset of B ($A \subset B$), and also a subset ($A \subseteq B$).
- A set B is a **superset** of set A if every element of A is also an element of B . This is written as $B \supseteq A$.
- A set B is a **proper superset** of set A if $B \supseteq A$ and $A \neq B$. This is written as $B \supset A$.
- **Example:** If $A = \{1\}$ and $B = \{1, 2, 3\}$, then B is a proper superset of A ($B \supset A$), and also a superset ($B \supseteq A$).
- If $A = B$, then A is an **improper subset** of B ($A \subseteq B$), and B is an **improper superset** of A ($B \supseteq A$).

6. Empty Set

- A set with no elements.
- Example: $S = \emptyset$ or $S = \{\}$
- Note: $\emptyset$ is a subset of any Set!

7. Disjoint

- Set A is Disjoint from B if there is no element of A in B .
- Example: if $A = \{1, 2\}$ and $B = \{4, 5\}$, then A and B are disjoint.

8. Singleton Set

- A set with exactly one element.
- Example: $S = \{1\}$

4 Notation and Operators

4.1 Ways to define a Set

1. Roster Form (Extensional Definition)

- This is the “basic” way of defining a Set
- Example: $S = \{1, 2, 3\}$
- Keep in mind the Set is defined by *extention*. Meaning numbers are “add” on.

2. Set-Builder Notation (Intensional Definition)

- A way of defining a set by specifying the properties that all its members must satisfy
- $S = \{X | X > 2\}$ or $S = \{X : X > 2\}$

4.2 Cardinality (#)

Cardinality is the amount of objects in a Set and is written like $\#(\text{Set Name})$

Example: $S = \{1, 2, 3\}$ so $\#(S) = 3$

4.3 Epsilon ($\in$)

The symbol $\in$ is used in set theory to denote whether an object is an element of a set.

- Example: $x \in \mathbb{N}$

This means that x is a natural number.

You can also use $\notin$ if an element is not in a set

4.4 Union ($\cup$)

$\cup$ is used to combine sets together

Example: $\{1, 2, 3\} \cup \{2, 3, 4, 5\} = \{1, 2, 3, 4, 5\}$

4.5 Intersection ($\cap$)

cap is used to find the same objects in sets

Example: $\{2, 3, 8\} \cap \{8, 3, 7, 5\} = \{8, 3\}$

4.6 Difference ($\setminus$)

$\setminus$ is used to find the elements of the first set that aren't in the second

$$A \setminus B = \{x \in A \mid x \notin B\}$$

5 Number Sets

Number sets are fundamental collections of numbers used in mathematics, each set containing specific types of numbers and nested within larger sets. The main number sets are:

- $\mathbb{N}$: **Natural numbers** (e.g., $0, 1, 2, 3, \dots$)
- $\mathbb{Z}$: **Integers** (e.g., $\dots, -2, -1, 0, 1, 2, \dots$)
- $\mathbb{Q}$: **Rational numbers** (numbers that can be written as fractions)
- $\mathbb{R}$: **Real numbers** (all rational and irrational numbers)
- $\mathbb{C}$: **Complex numbers** (numbers of the form $a + bi$)

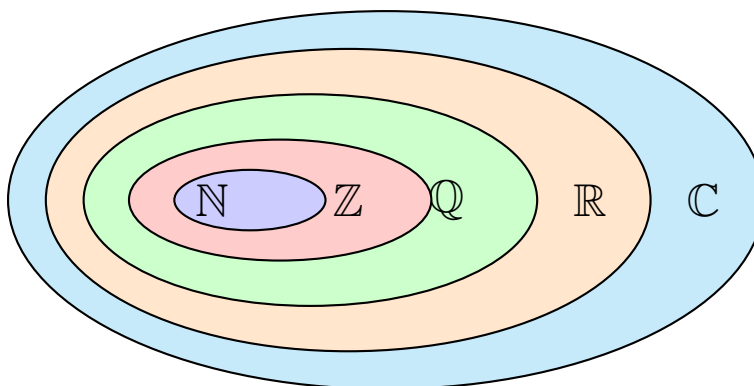


Figure 1: Order of the number sets

6 Completeness

6.1 Axiom of Completeness

Every nonempty set of real numbers that is bounded above has a least upper bound

Note: bounded above means there's an upper limit

6.2 Supremum and Infimum

Supremum is the least upper bound

For there to be a Supremum the following criteria must be met:

- There is an upper bound for the set
- sup must be less than all other upper bounds

Infimum is the greatest upper bound

This is the same as a Supremum just for the other “side”

Note: The Infimum or Supremum can either be inside or outside of the Set

7 Uniqueness

In mathematics, “unique” means that something is both one and only one. It’s the one and only object satisfying a specific condition

7.1 Proving Uniqueness

The most common way of proving uniqueness is by creating two entities a, b and if the answer is unique they must equal each other.

Example: $3 + 2 = 5$

To say that this is unique and there is no other solution set:

$$a + 2 = 5 \text{ and } b + 2 = 5$$

then because they must be equal to each other

$$a + 2 = b + 2$$

this cancel’s to

$$a = b$$

therefore

3 is a unique solution to $x + 2 = 5$